

Vladyslav Vashchuk

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EDUCATION

Susquehanna University — Selinsgrove, PA

Bachelor of Science in Computer Science

Expected May 2027

Department of Mathematics and Computer Sciences

Honors: *Dean's List* (Spring 2025, Fall 2025, Spring 2026)

GPA: 3.4 / 4.0

RESEARCH EXPERIENCE

Undergraduate Research Assistant – Machine Learning for Biomedical Signal Analysis

Department of Computer Science, Susquehanna University | Sep 2025 – January 2026

- Conducted research on automated ECG anomaly detection using the Random Cut Forest (RCF) algorithm for unsupervised time-series analysis.
- Applied signal preprocessing and feature extraction to analyze ECG waveform components (P wave, QRS complex, T wave).
- Implemented and evaluated models using Python, scikit-learn, NumPy, Pandas, and Matplotlib, measuring performance using accuracy, sensitivity, specificity, precision, recall, and F1-score.
- Designed a modular anomaly detection pipeline enabling scalable experimentation with machine learning models for biomedical signal processing.

PROJECTS

Growth OS — AI-Assisted Personal Development System | Founder & Developer

(Next.js, TypeScript, Supabase, PostgreSQL, Tailwind, AI APIs)

- Designed and built a full-stack personal development platform that structures goal planning, weekly reflections, and behavioral insight tracking across career, personal, and internal growth pillars.
- Architected a modular system using Next.js and Supabase with relational data models for goals, reflections, and adjustment logs to support structured journaling and long-term progress analysis.
- Implemented AI-assisted reflection analysis and pattern detection to surface behavioral trends, goal alignment insights, and structured feedback from user journaling data.
- Designed the platform architecture to support future AI coaching agents and longitudinal life analytics systems.

AI Claims Processing Copilot

Personal Project

- Developed an AI-powered claims triage and prioritization system simulating real insurance workflows
- Built a multi-claim dashboard with automated priority scoring, fraud signal detection, and workflow state management
- Implemented LLM-based claim analysis to extract structured data and generate explainable decision insights
- Designed decision-support features, including recommended actions and priority reasoning
- Demonstrated potential to reduce manual claims processing time by 30–50%

PROFESSIONAL EXPERIENCE

Operations & Project Support Associate

Quinn's Auction Galleries – Falls Church, VA | May 2024 – Present

- Support directors on warehouse workflow initiatives and operational process improvements, including development planning for FlowOps, a rack tracking and warehouse coordination platform designed to improve intake visibility and processing flow.
- Optimize warehouse layouts and transport routes to reduce operational bottlenecks and improve inventory movement.
- Help identify warehouse bottlenecks and workflow inefficiencies through hands-on operational analysis across intake, unpacking, lotting, and shipment preparation stages.
- Manage preparation of 40+ weekly shipments, overseeing picking, packing, labeling, and quality assurance.

RELEVANT COURSEWORK & TECHNICAL APPLICATIONS

Data Mining (*Python, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn*) — Spring 2025

- Implemented machine learning algorithms for classification, regression, clustering, and association rule mining on real-world datasets.
- Performed data preprocessing, feature engineering, and visualization to extract patterns and communicate insights.

Data Structures (*C, Python, Git, Visual Studio Code*) — Fall 2024

- Designed and implemented core data structures including linked lists, stacks, queues, trees, and graphs.
- Analyzed algorithm complexity and optimized searching and sorting techniques.

Software Engineering: Methodology (*Python, UML, GitHub, JIRA*) — Fall 2025

- Developed projects following the full Software Development Life Cycle (SDLC), including requirements, design, testing, and documentation.
- Applied object-oriented design, UML diagrams, and collaborative development tools (GitHub, JIRA).

Object-Oriented Programming & Design (*C++, Python, PyCharm, Git*) — Spring 2025

- Built modular programs using object-oriented concepts such as classes, inheritance, and polymorphism.
- Emphasized structured design, code readability, and optimization in multi-file applications.

Computer Organization (*Assembly, Logisim, Linux Terminal, Emu8086*) — Fall 2025

- Studied computer architecture, including digital logic, CPU registers, memory hierarchy, and OS interfaces.
- Implemented assembly programs to simulate arithmetic operations and low-level data processing.

LEADERSHIP & INVOLVEMENT

First-Year Experience Assistant — *Susquehanna University | Aug 2024 – Present*

- Guided new students on registration and academic resources.
- Supported 25+ residents with housing, orientation, and campus engagement.

Pi Kappa Phi – Iota Epsilon Chapter — *Member | Feb 2024 – Present*

- Assisted in organizing fundraising and service events for The Ability Experience charity.
- Helped coordinate logistics for events involving 200+ students and community participants.

TECHNICAL SKILLS

Programming: Python, C, C++, Assembly, SQL (basic), HTML/CSS

Machine Learning: scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, anomaly detection, time-series analysis

Development Tools: Git, GitHub, Jupyter Notebook, Google Colab, Visual Studio Code, PyCharm, UML

Systems & Infrastructure: Linux, modular software design, algorithm design, data pipelines

Languages: English (Professional), Ukrainian (Native), Russian (Native)